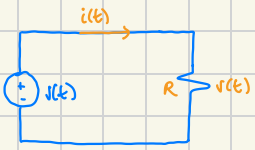
ELG 3125 - Signals & Systems

Lecture 2 - Part 1

Signal Energy and Power



$$p(t) = v(t)i(t) = v(t)\left[\frac{v(t)}{R}\right] = \frac{v^2(t)}{R}$$

$$p(t) = \frac{dE}{dt} \rightarrow dE = p(t)dt \rightarrow E_{total} = \int_{t_1}^{t_2} dE = \int_{t_1}^{t_2} p(t)dt$$

$$\rightarrow \boxed{E_{total} = \int_{t_1}^{t_2} p(t)dt}$$

$$E_{total} = \int_{t_1}^{t_2} p(t)dt = \int_{t_1}^{t_2} \frac{1}{R} v^2(t)dt$$

Average power:

$$\boxed{P_{ave} = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} p(t)dt = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} \frac{1}{R} v^2(t)dt}$$

For any continuous-time signal $x(t)$ → total energy in interval: $t_1 \leq t \leq t_2$ is: $\boxed{E_{total} = \int_{t_1}^{t_2} |x(t)|^2 dt}$

time-averaged power:

$$\boxed{P_{ave} = \frac{1}{t_2 - t_1}\int_{t_1}^{t_2} |x(t)|^2 dt}$$

For discrete-time signal $x[n]$ in $n_1 \leq n \leq n_2$ → $\boxed{E_{total} = \sum_{n_1}^{n_2} |x[n]|^2}$

Average power is: $\boxed{P_{ave} = \frac{1}{n_2 - n_1 + 1}\sum_{n_1}^{n_2} |x[n]|^2}$

If we have infinite time interval: $-\infty \leq t \leq \infty$ or $-\infty \leq n \leq \infty$

$$\boxed{E_\infty = \lim_{T\to\infty}\int_{-T}^{T} |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt}$$ continuous

$$\boxed{E_\infty = \lim_{N\to\infty}\sum_{-N}^{N} |x[n]|^2 = \sum_{+\infty}^{\infty} |x[n]|^2}$$ discrete

The time-averaged power over an infinite interval

$$\boxed{P_\infty = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} |x(t)|^2 dt}$$

$$\boxed{P_\infty = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{-N}^{N} |x[n]|^2}$$

Three classes of signals:

- Class 1: finite total energy ($E_\infty < \infty$), zero average power [Energy signal] → $P_\infty = \lim_{T\to\infty}\frac{E_\infty}{2T} = 0$
- Class 2: finite average power P_∞. For $P_\infty > 0$, $E_\infty = \infty$ [Power signal]
- Class 3: Neither P_∞ nor E_∞ finite.

Energy signal: total energy E_∞ is finite, $0 < E_\infty < \infty$, $P_\infty = 0$

Power signal: average power P_∞ is finite, $P_\infty < \infty$, $E_\infty = \infty$